(i) $\dfrac{dy}{dx} = u + x\dfrac{du}{dx}$

$$u + x\frac{du}{dx} = \frac{x}{ux} + \frac{ux}{x} = \frac{1}{u} + u$$

$$\implies u\frac{du}{dx} = \frac{1}{x}$$

$$\frac{1}{2}u^2 = \ln x + C$$

When $x = 1,\ y = 2,\ u = 2 \implies C = 4$

$$u^2 = 2\ln x + 4$$

$$y = ux = x\sqrt{2\ln x + 4}$$

$$\because 2\ln x + 4 > 0$$

$$\therefore x > e^{-2}$$

(ii) Let $y = ux,\ \dfrac{dy}{dx} = u + x\dfrac{du}{dx}$

$$u + x\frac{du}{dx} = \frac{x}{ux} + \frac{2ux}{x} = \frac{1}{u} + 2u$$

$$\frac{u}{1+u^2}\frac{du}{dx} = \frac{1}{x}$$

$$\frac{1}{2}\ln(1+u^2) = \ln x + C$$

$$1 + u^2 = Cx^2$$

When $x = 1,\ y = 2,\ u = 2 \implies C = 5$

$$1 + \frac{y^2}{x^2} = 5x^2$$

$$y^2 = 5x^4 - x^2 = x^2(5x^2 - 1)$$

$$y = x\sqrt{5x^2 - 1}, \quad x > \frac{1}{\sqrt{5}}$$

(iii) Let $y = ux^2,\ \dfrac{dy}{dx} = x^2\dfrac{du}{dx} + 2ux$

$$x^2\frac{du}{dx} + 2ux = \frac{x^2}{ux^2} + \frac{2ux^2}{x} = \frac{1}{u} + 2ux$$

$$u\frac{du}{dx} = \frac{1}{x^2}$$

$$\frac{1}{2}u^2 = -\frac{1}{x} + C$$

When $x = 1,\ y = 2,\ u = 2 \implies C = 6$

$$\frac{y^2}{x^4} = u^2 = 6 - \frac{2}{x}$$

$$y^2 = 6x^4 - 2x^3 = x^3(6x - 2)$$

$$y = x^{\frac{3}{2}}\sqrt{6x - 2}, \quad x > \frac{1}{3}$$

This solution was submitted by **Jon**

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